TEXORA · LECTURE SHEET

Fibres and Fabric Selection Criteria

How an engineer chooses: translating an application's demands into required properties, screening candidate fibres and structures against them, and deciding with numbers — illustrated by a fully worked strength-to-weight comparison.

TE-4206 · TECHNICAL TEXTILE AND NANO-TECHNOLOGY · TOPIC 3 · LEVEL 4 TERM 2

1. Selection is a procedure, not a preference

Topic 2 catalogued what fibres and structures *can* do. Selection is the disciplined reverse: start from the application and eliminate everything that cannot survive it. The procedure below is the one used in specification work:

1. **Define the duty.** What must the product do, where, for how long, and what happens if it fails? (A failed shade net and a failed lifting sling are not the same class of event.)
2. **List the service environment.** Temperature range, UV exposure, chemicals, moisture, abrasion, biological attack, cleaning method.
3. **Convert duty into measurable properties.** Required breaking force and elongation, permeability window, pore size, thermal resistance, flammability class, service life — each with a test method.
4. **Screen fibres by exclusion first.** Any fibre that fails a hard limit (melting point, chemical attack, UV life) is out, however attractive its other properties.
5. **Screen structures the same way.** The structure must be able to deliver the property set (e.g. a stable pore size excludes loose knits; thickness-direction strength demands 3D structures).
6. **Compare the survivors on the deciding ratio.** Strength-to-weight, performance per cost, life-cycle cost — with calculations, as in Section 4.
7. **Check processability and testing.** Can your machinery make it? Can the claim be tested and certified? An untestable specification is an opinion.

2. The property checklist

PROPERTY GROUP	ASK	TYPICAL DECIDING PROPERTIES
Mechanical	What loads, static or dynamic, in which directions?	Tenacity, modulus, elongation, tear and burst strength, abrasion, creep, fatigue

Thermal	Continuous and peak temperatures? Flame contact?	Melting/softening point, limiting oxygen index, char behaviour, thermal conductivity
Chemical / biological	Acids, alkalis, solvents, fuels, microbes, body fluids?	Chemical resistance, hydrolysis resistance, rot and mildew resistance
Environmental	Sunlight, weather, seawater, soil?	UV stability, wet-strength retention, soil burial behaviour
Fluid handling	Must it pass, hold, or wick a fluid?	Air/water permeability, pore size, absorbency, hydrostatic head
Dimensional	Must size stay fixed under load, heat, washing?	Shrinkage, thermal stability, creep, elastic recovery
Safety / regulatory	Which standards and certifications apply?	Flammability class, biocompatibility, food contact, protective-equipment standards
Economic	Cost per what — metre, or year of service?	Fibre price, processability, expected service life, replacement cost

Table 1 — The selection checklist. A candidate is only "good" relative to a completed row of this table for the specific application.

3. Rules of thumb that survive screening

- **Temperature excludes first.** Polypropylene (~165 °C melt) and HMPE (~150 °C) disappear from hot duties before any discussion of their excellent strength or price; polyester (~260 °C melt) serves moderate heat; aramids, glass, PPS and PTFE take the hot end.
- **Water is a property changer.** Polyamide loses strength wet and gains elongation; polyester and polypropylene barely notice water; cellulosics gain wet strength management problems in hygiene design but win on absorbency.
- **UV punishes the unprotected.** Para-aramid and HMPE need protection or cover in permanent outdoor exposure; polyester and acrylics are the weather-tolerant defaults.
- **Creep disqualifies quietly.** A fibre can pass every short test and still fail a ten-year static load by creep — decisive for ropes, geotextiles under permanent load, and suspended structures.
- **The weakest link is usually an interface.** Coatings delaminate, seams fail before fabric, fibres degrade where they are clamped. Selection includes joints and finishes, not only the headline fibre.

4. Worked example — strength-to-weight decides a rope

A mooring/utility rope must reach a breaking force of **50 kN**. Compare a steel wire rope with an HMPE fibre rope on material requirement. Required cross-sectional area A = F/σ, and mass per metre = A·ρ. Using typical textbook values — steel wire σ ≈ 1.77 GPa, ρ = 7,850 kg/m³; HMPE σ ≈ 3.5 GPa, ρ = 970 kg/m³:

WORKED EXAMPLE — 50 KN BREAKING FORCE

Steel: A = 50,000 / 1.77 × 10^9 = 2.82 × 10^{-5} m² = 28.2 mm². Mass per metre = 2.82 × 10^{-5} × 7,850 ≈ **0.222 kg/m**.

HMPE: A = 50,000 / 3.5 × 10^9 = 1.43 × 10^{-5} m² = 14.3 mm². Mass per metre = 1.43 × 10^{-5} × 970 ≈ **0.0139 kg/m**.

Ratio: 0.222 / 0.0139 ≈ **16** — the HMPE rope material is about sixteen times lighter for the same breaking force, and it floats (ρ < 1,000 kg/m³), which is itself a selection property at sea.

Cautions: these are solid-material figures — a real rope converts only part of its cross-section into strength (construction efficiency), and HMPE's creep and low melting point must still pass the Section 2 screening. The calculation decides the shortlist; the checklist decides the winner.

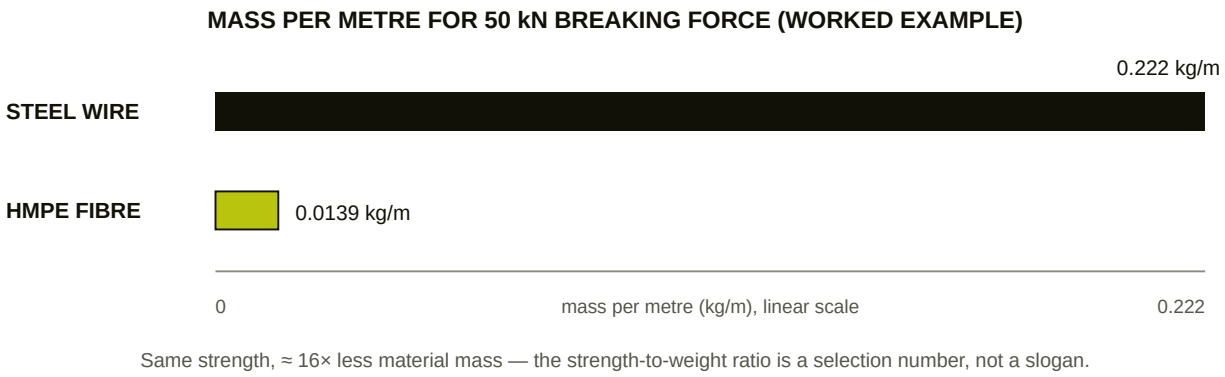


Figure 1 — Bar comparison drawn to scale from the worked example (steel bar length ∝ 0.222; HMPE bar ∝ 0.0139).

5. Cost is a ratio, not a price

High-performance fibres cost many times more per kilogram than commodity fibres, yet win applications — because the honest comparison is **cost per unit of delivered performance over service life**: cost per kN of breaking force, per year of filter life, per vehicle kilogram saved. Selection reports should state the ratio used. A cheaper product replaced three times in the life of a dearer one is usually the expensive choice, before counting the cost of the failures in between.

Glossary

Screening — eliminating candidates that fail any hard limit before comparing the rest. **Creep** — slow continuing extension under a sustained load. **Specific strength** — strength divided by density; the basis of strength-to-weight comparison. **Service life** — the period over which the product meets its specification in its actual environment. **Construction efficiency** — the fraction of total fibre strength realised in the made rope or fabric.

Key points

• Selection runs duty → environment → measurable properties → screen by exclusion → compare survivors on a deciding ratio → verify processability and testing.
• Temperature, water, UV and creep eliminate more candidates than strength ever does.
• Interfaces (seams, coatings, clamps) are part of the selection, not an afterthought.
• Compare cost per unit performance over service life, never price per kilogram alone.
• Every claim in a selection report needs a test method — untestable criteria are opinions.

Practice

1. A rope must break at 80 kN. Repeat Section 4 for steel (1.77 GPa, 7,850 kg/m³) and HMPE (3.5 GPa, 970 kg/m³) and give both masses per metre and the ratio.
2. An outdoor awning fabric must serve ten years in strong sun at up to 70 °C. Two candidates remain: HMPE and polyester. Apply the Section 2 checklist and choose, giving two deciding reasons.
3. A geotextile under a permanent embankment load passes all short-term tensile tests in PP and in PET. Which additional property must be checked before choosing, and why might it reverse the choice?

Answer key

1. Steel: $A = 80{,}000/1.77 \times 10^{9} = 4.52 \times 10^{-5}$ m²; mass $= 4.52 \times 10^{-5} \times 7{,}850 \approx$ **0.355 kg/m**. HMPE: $A = 80{,}000/3.5 \times 10^{9} = 2.29 \times 10^{-5}$ m²; mass ≈ **0.0222 kg/m**. Ratio ≈ **16** (the ratio is load-independent — it comes from the σ/ρ values).
2. Choose **polyester**. (1) UV: HMPE (like para-aramid) suffers significant UV degradation over long outdoor exposure; PET is among the most weather-tolerant commodity fibres. (2) Temperature margin and creep: 70 °C is far inside PET's comfort and a long way toward HMPE's softening region, and a tensioned awning is a permanent-load (creep) duty. HMPE's strength advantage does not survive this checklist.
3. **Creep** (long-term deformation under sustained load, tested as creep rupture / long-term design strength). PP creeps markedly more than PET; over a design life measured in decades the PP product

may need a much larger reduction factor, erasing its price advantage or failing the deformation limit — so the long-term property, not the short-term tensile result, decides.
